

PRESENTED BY :

Serene
Anouk
Claudia
Aren
Ladislav
Nina

IDE FUNDAMENTALS
02 October 2025

GROUP 9 PRESENTATION

PROBLEM STATEMENT



OBJECTIVE

How Might We encourage melatonin release in teenagers before bed to promote higher quality sleep?

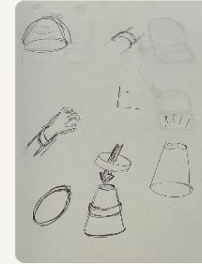
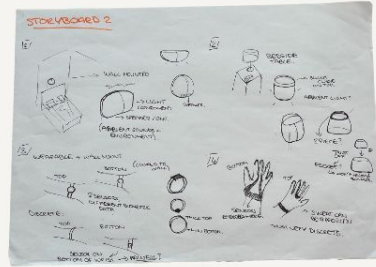
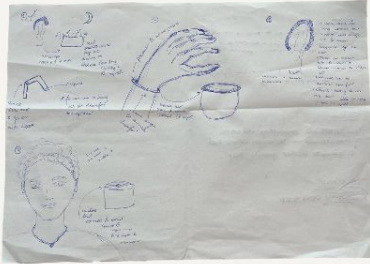
INSIGHTS

- **70–79%** of teens are on their phones 2 hours before bed.
- Blue light exposure from screens **delays melatonin** hormone signals.
- Sleep is important for the developing brain, and integral to **memory consolidation**.

HOW

- Increasing melatonin levels earlier in the night.
- Personalised sound therapy responsive to changes in the user's environment.
- Heightening memory recall by improving the quality of sleep.
- Building awareness of the importance of sleep health and routine.

THE DEVELOPMENT



SKETCH



PROTOTYPE



TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY TOUCH PLAY

THE ASSUMPTIONS



1

The algorithm that processes the biometric data will be able to analyse how sound impacts the users stress levels.



2

Teenagers will not engage with their phones immediately before bed and disrupt their sleep cycle.



3

Teenagers will consistently wear the product.

THE VIDEO



WEARABLE MECHANISMS



Heart Rate



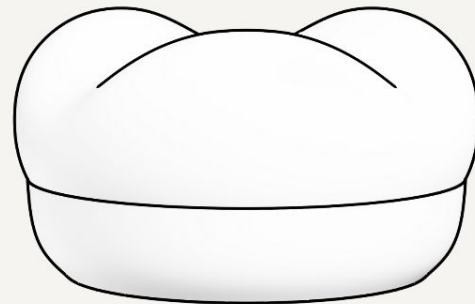
Blood Oxygen
Levels



Skin
Temperature

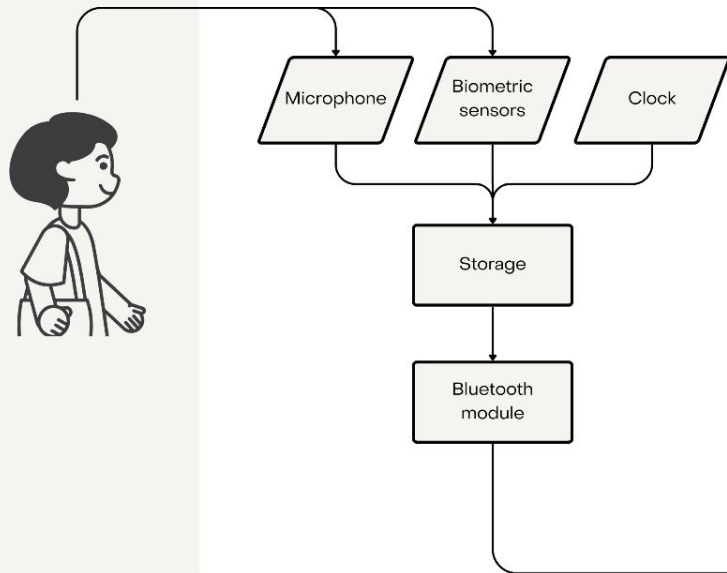


Humidity
(Sweat)

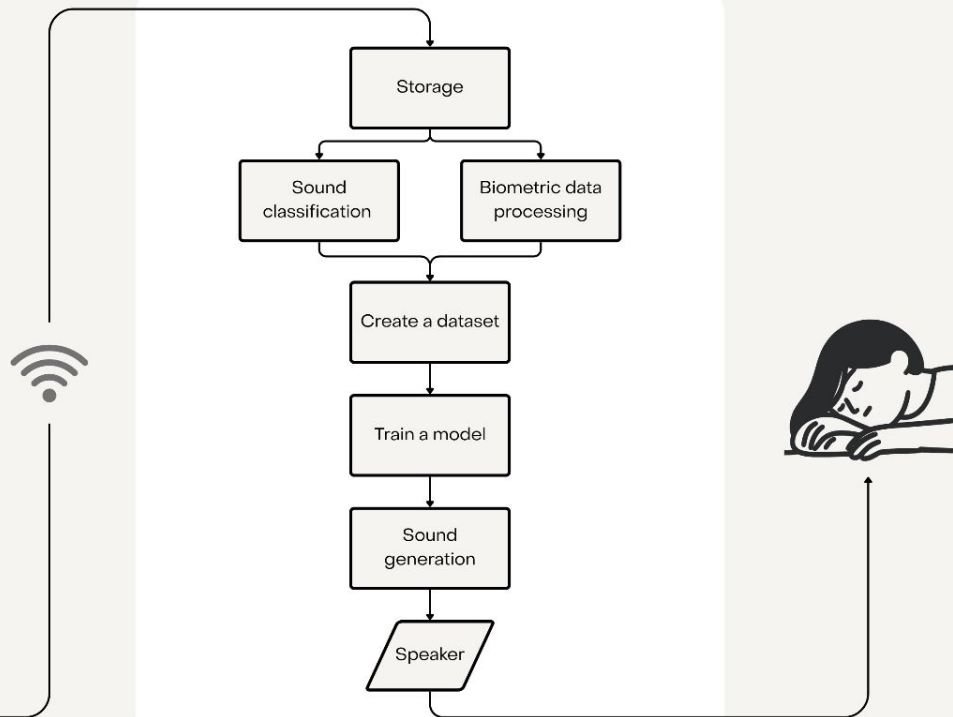


THE PIPELINE

Wearable Device



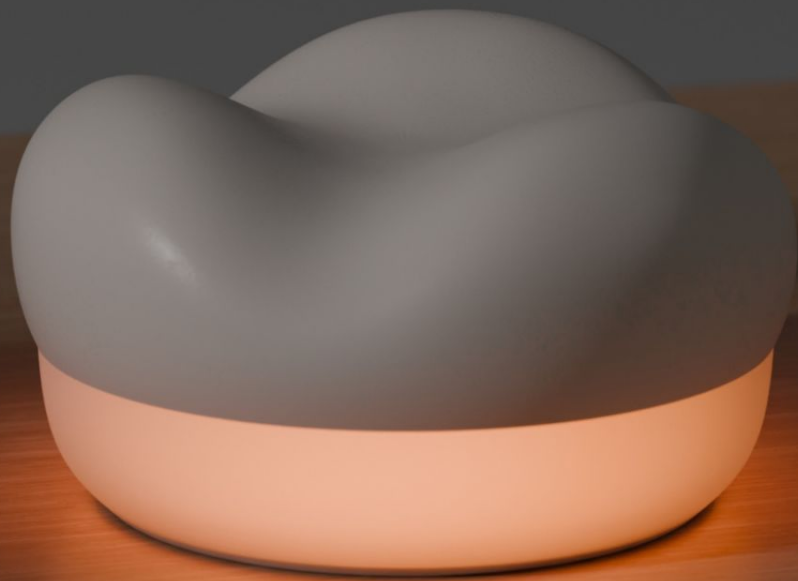
Home Device



DATA PROCESSING



SOMNIA
PROPOSAL



THANK
YOU!

Team 9

